

# Amit Singh Bhatti

## LINKS

[Portfolio Website](#), [Linkedin](#), [MS-Thesis](#), [Github](#), [Medium-Blog](#), [Substack](#)

## HIGHLIGHT

- Total 9+ YoE in AI solutions development comprising 2.5 YoE in leading Data Science teams.
- Top Talent in Reliance Jio R&D for two consecutive years with an A+ rating (2022-2024)
- Former co-founder and AI R&D Head of Sports Injury Management Startup(Dallas, US)
- NLP(Sentiment Information Mining) Data Science Hackathon Finalist - Runner-up

## SKILLS

**GenAI** : Deepseek Models, Qwen Models, Llama Models, GPT Models, Claude Models, OpenAI API, Langchain, LlamaIndex, Prompt Engineering, Mistral Models, BERT, T5, RAG, Vector Databases, LoRA, PEFT, Weaviate

**Agents** : LangGraph, LangChain, LlamaIndex, AutoGen/crewai, DSPy, Tool calling / function calling, MCP, A2A, ACP

**Data Science Libraries** : Pytorch, Tensorflow, Numpy, Pandas, HuggingFace, Scikit-Learn, FastText, Dask, NLTK, spaCY, Gensim, OpenCV, Matplotlib, Seaborn, Plotly, Bokeh, PySpark, Streamlit

**Deployment** : Docker, Flask, FastAPI, TorchServe, vLLM, TensorRT-LLM, ONNX, Prometheus, Triton Inference Server, Logstash, MLFlow, KubeFlow, Kubernetes, CUDA | FP16/BF16 | mixed precision

## EMPLOYMENT HISTORY

Feb 2025 — Present

AI Research Lead - Efficiency and Optimisation, Quantiphi R&D

Bengaluru

Research-to-product lead for LLM routing and agents workflow optimization powering customer support and finance assistants. Built novel IP filed SLA-aware routing framework that picks the right model and reasoning budget per query; standardized multi-agent orchestration (tools, retrieval, approvals) for cost-efficient resolution. With Agentic routing optimizations in place, the pipeline run cost reduced by factor 2.5x with SLMs and right tool selections.

Dec 2021 — Jan 2025

Senior AI Scientist, Reliance Jio

Hyderabad

- **Jio Telecommunication Query Resolution System - Led an NLP team of six AI engineers** to develop an in-house proprietary LLM-based query resolution engine in Indic Languages. The system Llama3.1-8B was instruction fine-tuned using SFT + LoRA with an in-house dataset of Hindi, Marathi and Telugu. Further, guided the team and contributed in optimization(Pruning+Quantization) and hosting of trained using vLLM inference server to bring down the inference cost by a factor of 12x.
- **Research Work** - Towards Open Domain Automatic Information Extraction in Hindi (AAAI Workshop) - Developed novel hybrid in-house Information Extraction system(Part of Speech tagging, Sense Disambiguation) with Llama2-70B + CRF decoder which was benchmarked with StanfordCoreNLP, Llama2 70B, and OpenAI API-4o. Achieved significant performance gains of 2.5% above baseline metrics through systematic improvements.

Sept 2020 — Nov 2021

Data Scientist, Legato Health Technologies - Anthem Inc

Hyderabad

- Led development of an end-to-end meeting transcription system using NVIDIA NeMo toolkit, implementing multi-speaker diarization with TitaNet embeddings and MSDD (Multi-Scale Diarization Decoder), achieving 95% speaker identification accuracy for medical conversations across diverse accents.
- Architected comprehensive NLP pipeline processing diarized text through fine-tuned transformers (BERT for sentiment/NER, T5 for summarization), automating medical Minutes of Meeting generation and reducing documentation time by 70%

Jan 2019 — Aug 2020

Deep Learning Software Engineer, Inspirit-IoT Technologies

Chennai

- Collaborated with UIUC PhD researchers to develop DNNOptimizer, a proprietary neural network compression framework that surpassed TensorRT benchmarks for vision models (ResNet-50, YOLOv3, FaceNet) on Intel Arria FPGAs
- Implemented and optimized object detection and facial recognition pipelines through hardware-aware compression techniques, working closely with architecture experts for FPGA-specific optimizations

- Pioneered deep learning research for insurance claims classification using hybrid LSTM/GRU models with custom text preprocessing, achieving 87% accuracy; designed insurance-specific embeddings and hierarchical classification improving model interpretability by 40% and reducing processing errors by 25%
- Architected production ETL pipeline handling 100K+ daily claims, contributing to a classification patent and implementing domain-specific NLP models (SVM, LSTM, GRU); integrated real-time fraud detection achieving 92% accuracy across multiple insurance categories; designed scalable microservices architecture reducing processing latency by 60% while maintaining 95% uptime

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PATENTS

Triad Three Coordinate based degree of turns and frequency of turns computation for kinematics sports movement - Patent Number(India) - 202311036108

Kinematics Spatial Vision Grading System for Prediction of Anterior Cruciate Tears - Patent Number(India) - 202311036107

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EDUCATION

Oct 2020 — Jul 2022

MS by Research (NLP) - Data Science, Liverpool John Moores University

Thesis: Neuroticism Alignment Detection for Misogyny Aggression via Hierarchical Explanation Method. Analyzed the HEDGE algorithm’s explanation weights for alignment with neuroticism, a key trait in Incel behavior.

Jul 2012 — Jul 2016

B.Tech - Robotics and Control Engineering, NIT Jalandhar